

1. Import libraries

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from IPython.display import display

pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)
pd.set_option("display.float_format", "{:.2f}".format)

plt.style.use("ggplot")
sns.set_theme(style="whitegrid")
```

[1]

```
import matplotlib.gridspec as gridspec
from IPython.display import Markdown

# Consistent color palette used across every chart
COLORS = {
    "Fatal": "#C0392B", "Serious": "#E67E22", "Slight": "#2980B9",
    "primary": "#2C3E50", "accent": "#16A085", "muted": "#95A5A6",
}
SEVERITY_ORDER = ["Fatal", "Serious", "Slight"]

def style_ax(ax, title):
    ax.set_title(title, fontsize=13, fontweight="bold", color=COLORS["primary"], pad=12)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.grid(axis="y", alpha=0.3)
```

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2. Load the data

```
# Load Datasets

collisions_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\collision.csv",
    low_memory=False
)

vehicles_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\vehicle.csv",
    low_memory=False
)

casualties_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\casualty.csv",
    low_memory=False
)

print("Collisions :", collisions_df.shape)
print("Vehicles   :", vehicles_df.shape)
print("Casualties  :", casualties_df.shape)
```

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```
... Collisions : (503475, 44)
    Vehicles   : (920692, 32)
    Casualties : (640522, 23)
```

Collisions													
	collision_index	collision_year	collision_ref_no	location_easting_osgr	location_northing_osgr	longitude	latitude	police_force	collision_severity	number_of_vehicles	number_of_casualties	date	day_of_week
0	2021170H10421	2021	170H10421	447098.00	532997.00	-1.27	54.69	17	3	2	2	22/05/2021	7
1	2021170H11231	2021	170H11231	450486.00	533118.00	-1.22	54.69	17	3	2	1	20/10/2021	4
2	2020170M11750	2020	170M11750	449694.00	519733.00	-1.23	54.57	17	3	2	1	01/12/2020	3
3	2021170M31761	2021	170M31761	449744.00	514217.00	-1.23	54.52	17	3	1	1	09/12/2021	5
4	2021170S10441	2021	170S10441	445971.00	520834.00	-1.29	54.58	17	3	3	1	12/04/2021	2
Vehicles													
	collision_index	collision_year	collision_ref_no	vehicle_reference	vehicle_type	towing_and_articulation	vehicle_manoeuvre_historic	vehicle_manoeuvre	vehicle_direction_from	vehicle_direction_to	vehicle_location_rest		
0	2020210979534	2020	210979534	2	5	0		3	3	4	6		
1	2020030959800	2020	030959800	1	5	0		17	19	2	6		
2	2020360978858	2020	360978858	2	5	0		18	19	8	4		
3	2020430340711	2020	430340711	1	5	0		18	19	3	7		
4	2020010260297	2020	010260297	1	5	0		99	99	9	9		
Casualties													
	collision_index	collision_year	collision_ref_no	vehicle_reference	casualty_reference	casualty_class	sex_of_casualty	age_of_casualty	age_band_of_casualty	casualty_severity	pedestrian_location	pedestrian_movement	
0	2020010280094	2020	010280094	1	1	3	2	24	5	3	8		5
1	202031D109620	2020	31D109620	1	1	3	2	95	11	2	5		1
2	2020401003715	2020	401003715	1	1	3	1	39	7	3	5		1
3	2021201086330	2021	201086330	1	1	3	1	63	9	3	6		5
4	2021371050963	2021	371050963	1	1	3	1	51	8	3	9		5

Missing Values

```
print("Collisions")
display(collisions_df.isnull().sum())
```

```
print("\nVehicles")
display(vehicles_df.isnull().sum())
```

```
print("\nCasualties")
display(casualties_df.isnull().sum())
```

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... Collisions

```
... collision_index      0
collision_year          0
collision_ref_no        0
location_easting_osgr   65
location_northing_osgr  65
longitude               65
latitude               65
police_force            0
collision_severity      0
number_of_vehicles      0
number_of_casualties    0
date                   0
day_of_week            0
time                   0
local_authority_district 0
local_authority_ons_district 0
local_authority_highway 0
local_authority_highway_current 80
first_road_class        0
first_road_number       0
road_type               0
speed_limit             0
junction_detail_historic 0
junction_detail         0
junction_control        0
second_road_class       0
second_road_number      0
pedestrian_crossing_human_control_historic 0
pedestrian_crossing_physical_facilities_historic 0
pedestrian_crossing     0
```

Road Safety Data Analysis (UK)

Business Problem

Road traffic accidents are a major public safety concern. By analyzing historical road safety data, we can identify accident patterns, high-risk conditions, and factors contributing to accident severity.

Objectives

- Explore road accident trends.
 - Analyze accident severity.
 - Investigate vehicle and casualty characteristics.
 - Identify temporal and environmental patterns.
 - Generate insights to support road safety improvements.
-

Dataset Description

This project uses the UK Road Safety datasets:

- **Collision Dataset:** Information about each accident.
- **Vehicle Dataset:** Information about vehicles involved.
- **Casualty Dataset:** Information about casualties involved.

Each dataset is linked using **accident_index**.

```
collisions_df[
    [
        "collision_severity",
        "weather_conditions",
        "light_conditions",
        "road_surface_conditions",
        "urban_or_rural_area"
    ]
].head()
```

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...

	collision_severity	weather_conditions	light_conditions	road_surface_conditions	urban_or_rural_area
0	Slight	Fine	Dark - No street lighting	Dry	Rural
1	Slight	Fine	Daylight	Dry	Urban
2	Slight	Fine	Dark - Street lights lit	Dry	Urban
3	Slight	Fine	Dark - Street lights lit	Wet or Damp	Rural
4	Slight	Fine	Daylight	Dry	Rural

✓ 7. Exploratory Data Analysis (EDA)

> 7.1 Distribution of Collision Severity

▷ 2 cells hidden ...

> 7.2 Trends Over Time

▷ 6 cells hidden ...

> 7.3 Environmental & Road Conditions

▷ 5 cells hidden ...

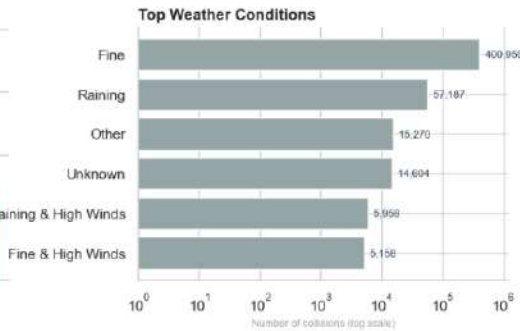
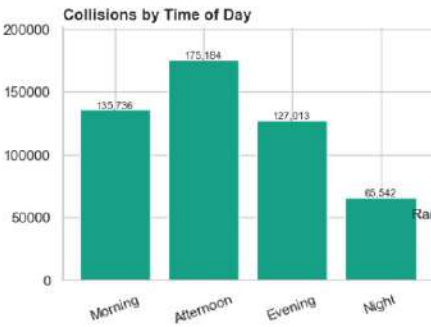
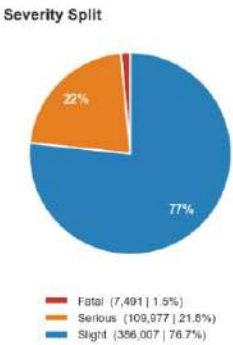
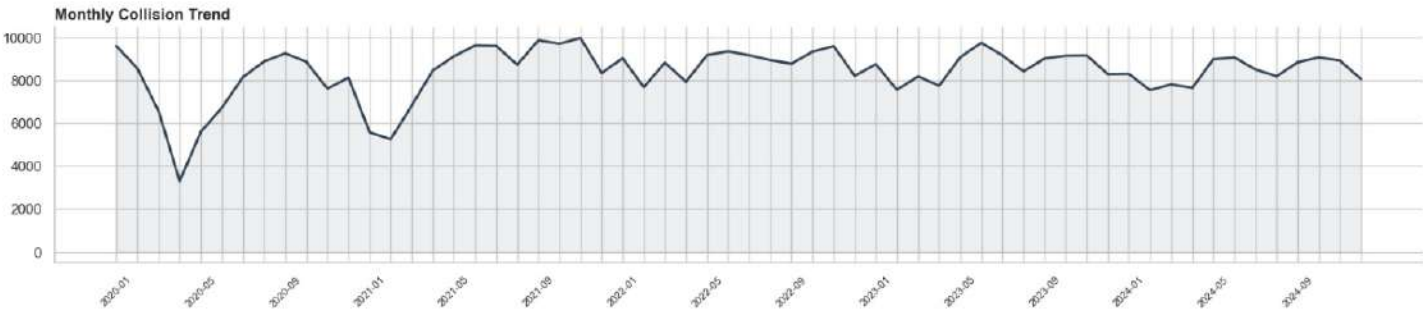
> 7.4 Severity vs. Contributing Factors

▷ 5 cells hidden ...

> 7.5 Casualties & Correlation

▷ 3 cells hidden ...

UK Road Safety — Executive Dashboard



5. Feature Engineering

```
# Extract Features from Date
```

```
collisions_df["year"] = collisions_df["date"].dt.year
collisions_df["month"] = collisions_df["date"].dt.month
collisions_df["month_name"] = collisions_df["date"].dt.month_name()

collisions_df["day"] = collisions_df["date"].dt.day
collisions_df["day_name"] = collisions_df["date"].dt.day_name()

print(collisions_df[
    ["date", "year", "month", "month_name", "day", "day_name"]
].head())
```

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```
...      date  year  month month_name  day  day_name
0 2021-05-22  2021     5      May    22  Saturday
1 2021-10-20  2021    10   October    20  Wednesday
2 2020-12-01  2020    12  December     1   Tuesday
3 2021-12-09  2021    12  December     9  Thursday
4 2021-04-12  2021     4   April    12   Monday
```

```
# Extract Hour
```

```
collisions_df["hour"] = pd.to_datetime(
    collisions_df["time"].astype(str)
).dt.hour

print(collisions_df[["time", "hour"]].head())
```

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```
...      time  hour
0 22:44:00    22
1 15:50:00    15
2 18:00:00    18
3 16:55:00    16
4 09:02:00     9
```